AI-POWERED FOOD RECOMMENDATION SYSTEM

SMART FOOD SUGGETION



Bachelor ofScience (Software Engineering)

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Project

Submitted in Partial Fulfillment

of the Requirements for the
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by

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DECLARATION

It is declared that this is an original piece of my own work, except where otherwise acknowledged in text and references. This work has not been submitted in any form for another degree or diploma at any university or other institution for tertiary education and shall not be submitted by me in future for obtaining any degree from this or any other University or Institution. I am the responsible if I do not meet the deadline.

Naveed Abbas Shah Signature: _____

2022S-MULBSSWE-021



DEDICATION

To

My Mother

Who always wished or prayed for my success, for supporting and encouraging me to believe in myself, who always sacrificed so much for me.

My Father

For earning an honest living for me,forsupporting and encouraging me to believeinmyself.

My Teacher

For his support and wishes.

CERTIFICATE OF APPROVAL

It is certified that the project titled "Multiplayer Battle Arena" carried out by Haider Raza, 2019f-mulbscs-201, under the supervision of Mr. Muhammad Kamran, Minhaj University Lahore, is fully adequate, in scope and in quality, as a final year project for the degree of BS of Computer Science.

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ABSTRACT

A game introduces a multiplayer shooting game designed to provide an exhilarating and immersive gaming experience. Players engage in first-paced battles against opponents from around the world, aiming to showcase their shooting skills and strategic prowess. The game features a diverse array of weapons, dynamic map, and engaging gameplay mechanics to keep players engaged and challenged throughout their gaming sessions. The implementation leverages cutting-edge technologies and networking protocols to ensure seamless online multiplayer functionality, enabling players to connect and compete in real time. Through comprehensive testing and evaluation, the game demonstrates its ability to deliver thrilling multiplayer action, fostering an industry by offering an exciting multiplayer shooting game that combines skill-based gameplay, immersive visuals, and an engaging social experience. "**Multiplayer Battle Arena**" is an intense online multiplayer shooting game that immerses players in adrenaline-fueled combat within immersive virtual environments. With a focus on skillful gunplay, strategic decision-making, and competitive multiplayer action, players engage in thrilling battles against opponents from around the world. With a wide range of game modes to choose from, players can team up with friends or compete individually in diverse gameplay scenarios. Whether it's customizing loadouts with an extensive arsenal of weapons and equipment, coordinating with teammates for strategic maneuvers, or adapting tactics to dynamic maps, "Multiplayer Battle Arena" delivers an unforgettable multiplayer shooting experience that will test players' skills and keep them engaged in the virtual battlefield.

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CHAPTER 1

INTRODUCTION

Basically, in this chapter we introducing the main steps of our project. This introduction contains the background, Goals/Objectives, Solutions, Gap Analysis, Gantt chart, Project Plan, Report outline. This material of introduction helps to understand the main issues related to this project.

1.1. Background

Online shooting games have become incredibly popular, allowing players worldwide to compete against each other in virtual combat. They originated in the 1990s with games like "Doom" and "Quake" and have since evolved with improved graphics, larger battles, and diverse gameplay mechanics. Online platforms and cross-platform play have facilitated connectivity among players, and future advancements in technology promise even more immersive experiences.

1.2. Goals and Objectives

The objective of online shooting games is to engage in multiplayer battles, defeat opponents, and secure victory. These games encourage teamwork, offer progression and rewards, cater to esports competitions, facilitate social interaction, and provide an immersive and enjoyable gaming experience.

1.3. Gap Analysis

Now a day there are many trending shooting games with a number of traffic of users but some of the features of these games are difficult to understand by some users. **Multiplayer Battle Arena** provides a user-friendly interface in which the features are very easy to understand., So, features of this game are better than other games in which user can easily play the game without any delay and enjoy the mechanism of the game without any restrictions.

1.4. Project Plan

Details: In this game user make a profile by providing a unique username, and after that he/she is able to play the game with the multiple players in the world wide.

Features: The different type of features all over the game will be up to date and new features will implement regularly.

1.4.1. Work Breakdown Structure

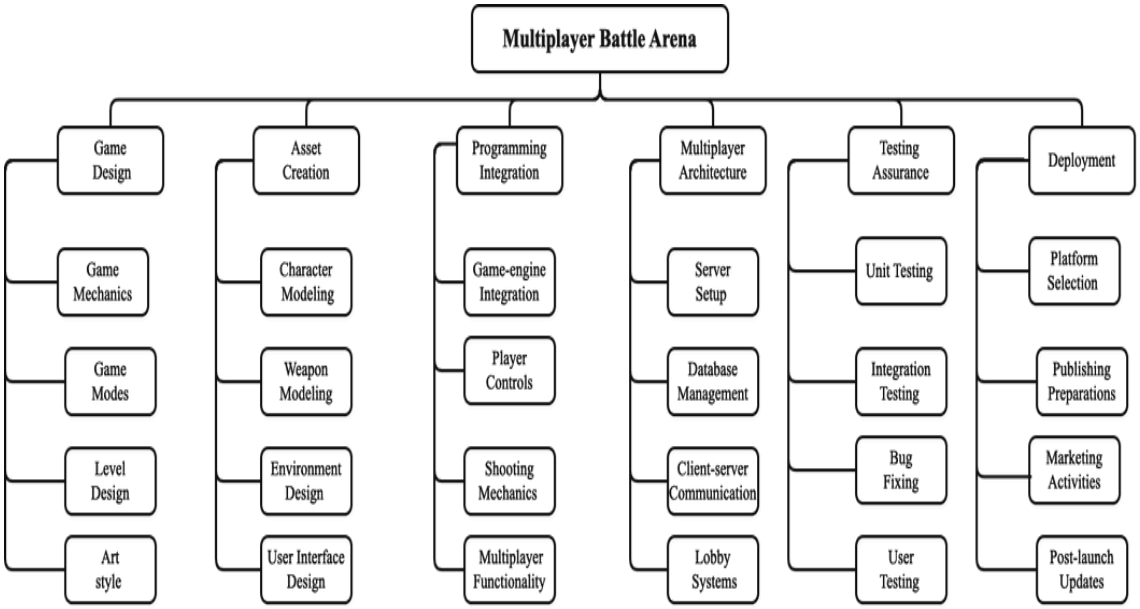


Figure 1.1 Word Breakdown Structure

1.4.2. Gantt Chart

Tasks to be performed	October 2022	November 2022	January 2023	February 2023	March-April 2023	May 2023	June 2023	July 2023
Pre analysis								
Project proposal, Meeting with supervisor about FYP idea								
Planning, thinking about game story, levels and learning technology								
Construct SRS document, choose tools and environment								
Start designing and implementation								
Developing , Testing and enhancement running with writing the report								
Documentation								
Final submission								

Figure 1.2 Gantt Chart

1.4.3. Team Members

Team consists of

• Project Manager (PM)

• Requirement Analyst (RA)

- Backend Developers
- Frontend developers
- Testing Engineering (TE)

Each member has some specific skills which they used efficiently in project. A table in given to show team members names, their skills and their task description.

Table 1.1 Team Member

Name	Skills	Description
Haider Raza	Back-end Developers Front-end Developer	Involved in coding Design The User Interface

1.5. Report Outline

Multiplayer Battle Arena provides a friendly user interface to the users. All the features are easy to understand.

• Game modes: Team Deathmatch, Free for All, and Capture of Flag.

• Player team system.

- Shooter AI (Bots).
- 6 different types of weapons: Rifle, Sniper, Shotgun, Pistol, Grenade, and Knife.

• Walk, Slide, Jump, Crouch, Shoot, Die, and Run player animations.

CHAPTER 2

SYSTEM REQUIREMENTS AND SPECIFICATION

2.1. Purpose

The purpose of an online shooting game is to provide an exhilarating and immersive gaming experience. Players engage in virtual combat, showcasing their skills, strategic thinking, and teamwork. These games offer entertainment and escapism, allowing players to enter thrilling virtual worlds. They foster social interaction by enabling players to team up with friends and join communities. With intense gameplay and a sense of achievement, online shooting games aim to deliver adrenaline-pumping action. The ultimate goal is to create an enjoyable and memorable experience that keeps players engaged and coming back for more.

2.1.1 Document Conventions

When writing the SRS document for "Multiplayer Battle Arena" the following terminologies are used to make the document more effective and readable we used font:

• Times New Roman

- Line spacing: 2 line spacing
- Font Size: 12pt
- headings: 16pt
- Headings are bold.

This document is intended to the audience and also for those End Users who really want to know for what purpose this game is developed for.

2.1.2. Intended Audience

People of all age groups owing a computer system or laptop.

2.2. Overall Description

2.2.1. Service Perspective

The online 3d multiplayer shooting game is developed to help people communicate during the game with others easily. The game is developed using c sharp as programming language and design and coded in Unity 3D.

2.2.2. Service Function

The main function of this project is the multiplayer matchmaking and player communicating. These functions enhance the gaming experience by facilitating fair match ups and enabling collaboration among players.

2.2.3. Product Functions

Features

- Login (as guest)
- Matchmaking
- Chat Communication
- Multiple play modes
- Spectator Mode
- Graphics Settings

User Function

- Login
- Select Mode
- Edit Settings
- Play Game-play

2.2.4. User Classes and Characteristics

Admin

The administrator can add, edit or delete user's accounts. Admin manages all functionality of Multiplayer Battle Arena.

Design and Implementation Constraints:

- The device will have 8GB RAM.
- Good quality Internet connection required.
- Operating System Windows or Macintosh.

Tools

- Unity 3D
- Photon Unity Networking
- C#

2.2.5. Assumptions and Dependencies

In this game, the user must login as a guest. User must have a computer or laptop and at least 4 GB Ram in system. Also, the user must have internet access because Multiplayer Battle Arena is an online game.

2.3. External Interface Requirements

2.3.1. User Interfaces

The game provides a user-friendly and interactive user interface. In this game, the user will easily understand the use of every element and enjoy the user experience.

User can interact with following things:

• User can play the game in multiple modes.

• Chat with your team members while in the match.

2.3.2. Hardware Interface

- Processor: i3
- Hard Disk: 5 GB
- Memory: 4 GB RAM
- Laptop or computer.

2.3.3. Game Interface

- Unity 3D
- Photon Unity Networking
- C#

2.4. System Features

2.4.1. User Login

Description Priority

User Login has a priority value. Whenever the user installs the game and working for the first time, he/she has to get login first as a guest

Stimulus/Response Sequences

- The game launched from the login screen.
- User is prompted to main menu.

Functional Requirements

User gets registered and is able to use the game.

2.4.2. Logout

Description and Priority

User has to authenticate if he/she is not logged in. He/she can also log out if wanted. This feature has 2nd highest priority at most.

Stimulus/Response Sequences

- The game launched from the login screen.
- User is prompted to main menu.
- Multiple options show to play the game.
- If the user is logging out, simply user back to login screen.

Functional Requirements

User gets logged in/ logged out

2.4.3. Edit Settings

Description and Priority

This feature is specifically for those users who want to change the settings according to their systems.

Stimulus/Response Sequences

- The user chooses settings option from main menu.
- Change settings according to his requirements.

Functional Requirements

User must be logged in and stay connected with internet.

2.4.4. Play Online

Description and Priority
This feature is specifically for those users who want to play game online.
Stimulus/Response Sequences

- The user can play the game with online people.
- User can play the game in 3 different modes.

Functional Requirements
User must be logged in and stay connected with internet.

2.5. Other Nonfunctional Requirements

2.5.1. Performance Requirements

- System device should have Window or Macintosh operating system.
- User must have a good internet connection to really enjoy the performance of game.
- Size of the game should be small that it can be accessible in every system if system is low storage device.

2.5.2. Safety Requirements

Only authorized member or user can log in into a game. Data should be saved in encrypted form so that in case of attack or hack data will be unreadable or unacceptable.

2.5.3. Software Quality Attributes

Privacy and security of the user is our basic and most important motive.

Availability

Users can access their accounts anytime anywhere.

Integrity

Better component design to get better performance at peak time.

Correctness

Game is design in that way that it should be free from errors.

Reliability

Game is reliable and the user can trust that it's 100% reliable.

Security

Game is 100% secure and users save their data with no worry. Before storing data on a database, data is encrypted so that no one can access or steal user's personal information.

Easy to Use

Game is easy to use and learn.

2.6. Other Requirements

The system must store the entire User information. For each user data saved with its unique id in the database.

CHAPTER 3

USECASE ANALYSIS

3.1. Use Case Model

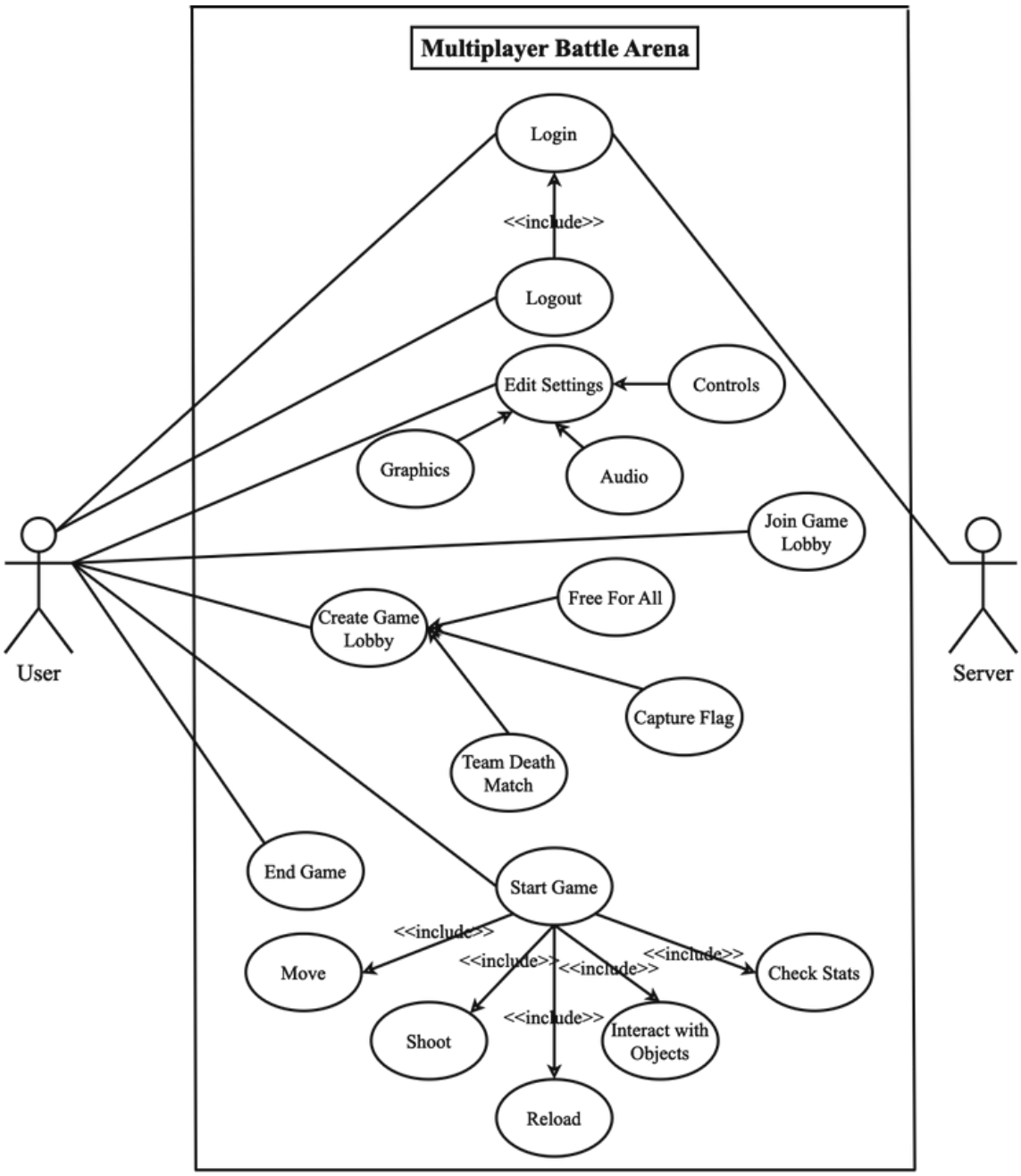


Figure 3.1 Use Case Model

3.2. Fully Dressed Use Cases

- **Login**

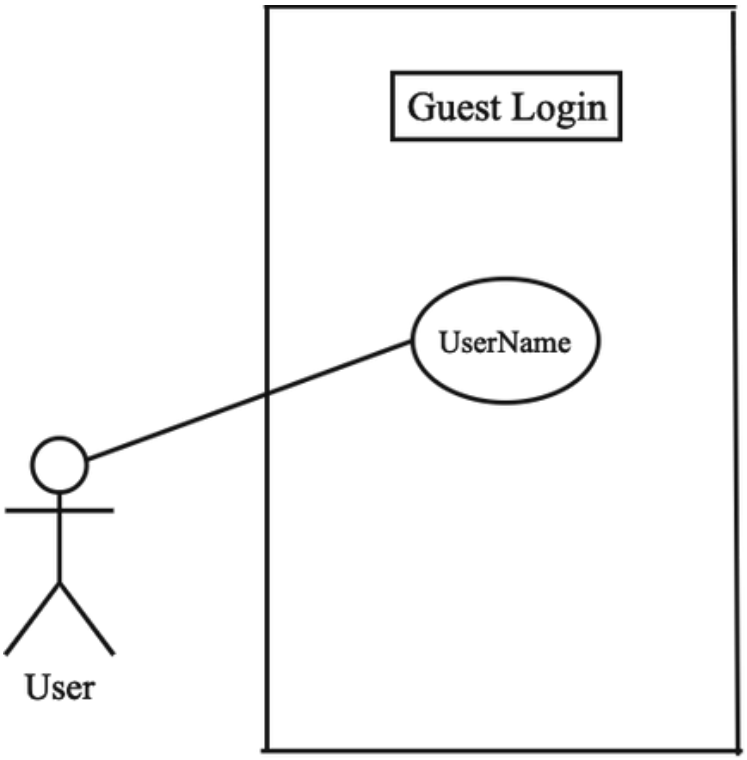


Figure 3.2 Login Use Case

Table 3.1 Use Case<01>

Fully Dressed Use Case		
•	**Use Case Name**	Login
•	**Scope**	Login
•	**Level**	User goal
•	**Actor**	User, Server
•	**Stakeholder**	User, Server
•	**Pre-Condition**	Internet Connection
•	**Post-Condition**	Access to game
•	**Success Scenario**	Successfully Login

- **Edit Settings**

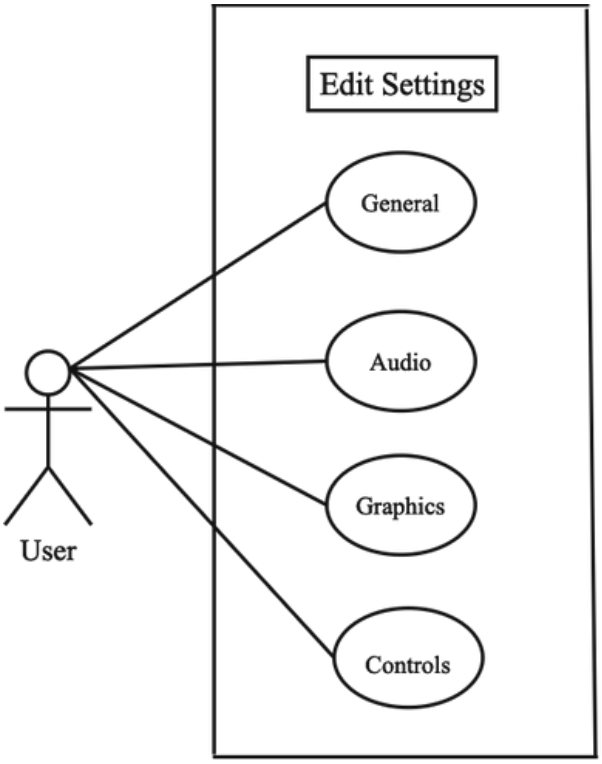


Figure 3.3 Edit Profile Use Case

Table 3.2 Use Case<02>

Fully Dressed Use Case		
•	**Use Case Name**	Edit Setting (General, Graphics, Audio, Control)
•	**Scope**	Settings
•	**Level**	User Goal
•	**Actor**	User
•	**Stakeholder**	User
•	**Pre-Condition**	Internet connection Must be login
•	**Post-Condition**	Settings should be edit
•	**Success Scenario**	Settings edit successfully

- **Join Game Lobby**

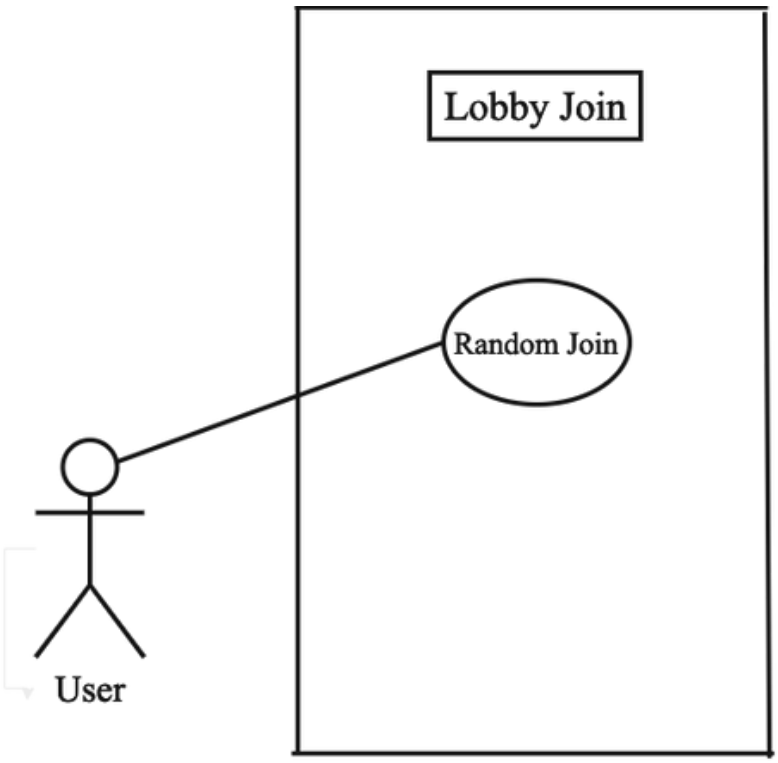


Figure 3.4 Join Game Lobby Use Case

Table 3.3 Use Case<03>

Fully Dressed Use Case		
•	**Use Case Name**	Join Game Lobby
•	**Scope**	Lobby
•	**Level**	User Goal
•	**Actor**	User
•	**Stakeholder**	User
•	**Pre-Condition**	Must be Login Internet Connection
•	**Post-Condition**	Join any lobby
•	**Success Scenario**	Lobby joined successfully.

- **Create Game Lobby**

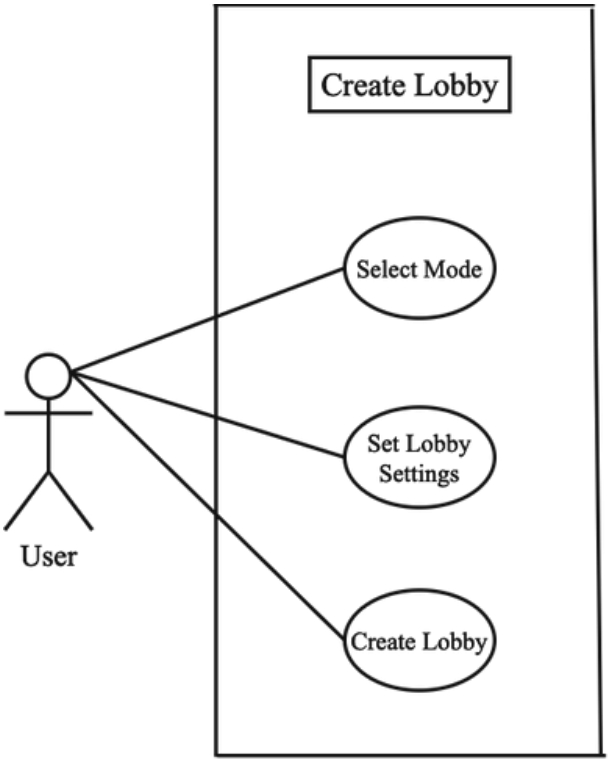


Figure 3.5 Create Lobby Use Case

Table 3.4 Use Case<04>

Fully Dressed Use Case		
•	**Use Case Name**	Create Game Lobby
•	**Scope**	Lobby
•	**Level**	User Goal
•	**Actor**	User
•	**Stakeholder**	User
•	**Pre-Condition**	Must be Login Internet Connection User must have to select mode User must set all settings for the mode
•	**Post-Condition**	Create a new lobby
•	**Success Scenario**	Lobby created Successfully

- **Start Game**

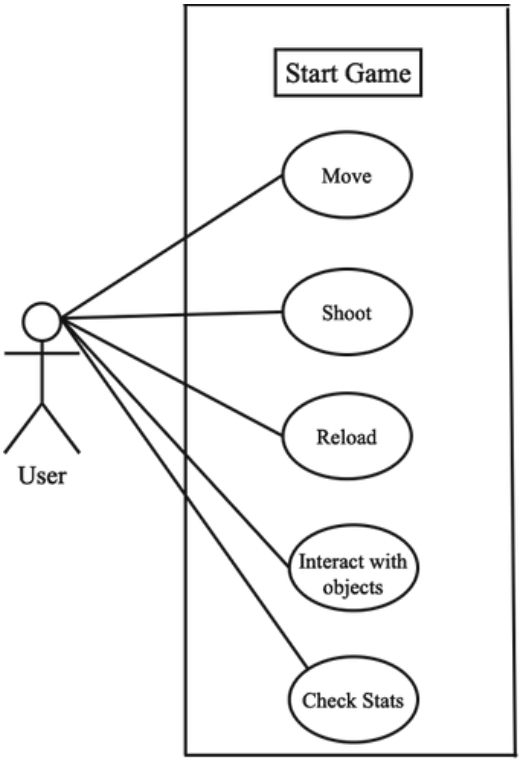


Figure 3.6 Start Game Use Case

Table 3.5 Use Case<05>

Fully Dressed Use Case		
•	**Use Case Name**	Start Game
•	**Scope**	Game
•	**Level**	User Goal
•	**Actor**	User
•	**Stakeholder**	User
•	**Pre-Condition**	Must be Login Internet Connection User must be join/create lobby
•	**Post-Condition**	Start Game
•	**Success Scenario**	Game started successfully

- **End Game**

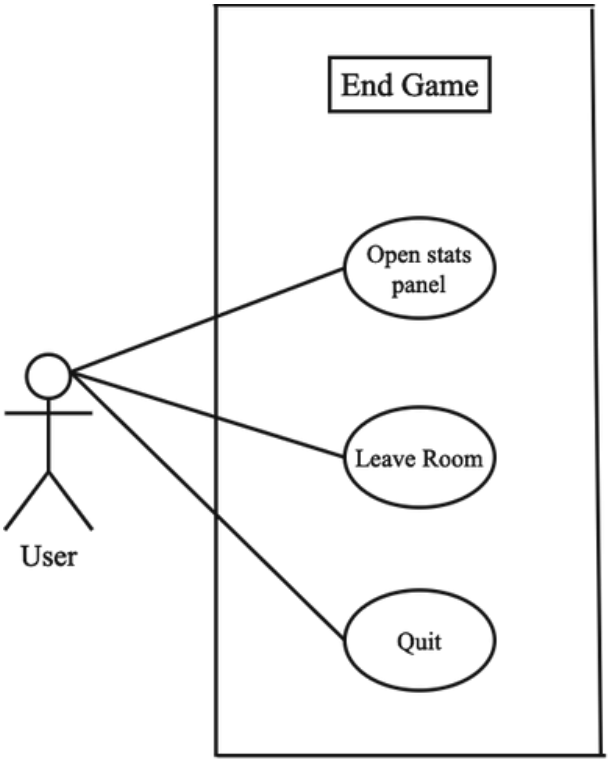


Figure 3.7 End Game Use Case

Table 3.6 Use Case<06>

Fully Dressed Use Case		
•	**Use Case Name**	End Game
•	**Scope**	Game
•	**Level**	User Goal
•	**Actor**	User
•	**Stakeholder**	User
•	**Pre-Condition**	Must be Login Internet Connection
•	**Post-Condition**	Game End
•	**Success Scenario**	Game Ended Successfully

CHAPTER 4

SYSTEM DESIGN

System Design

The system design describes the system requirement, operating environment system and subsystem architecture, files and database design, input format, output layout, user interface, detailed design processing logic, an external interface.

4.1. Architecture Diagram

We divided system architecture in three-part Game, services & authentication and Photon Unity Networking (PUN), As Shown in the figure.

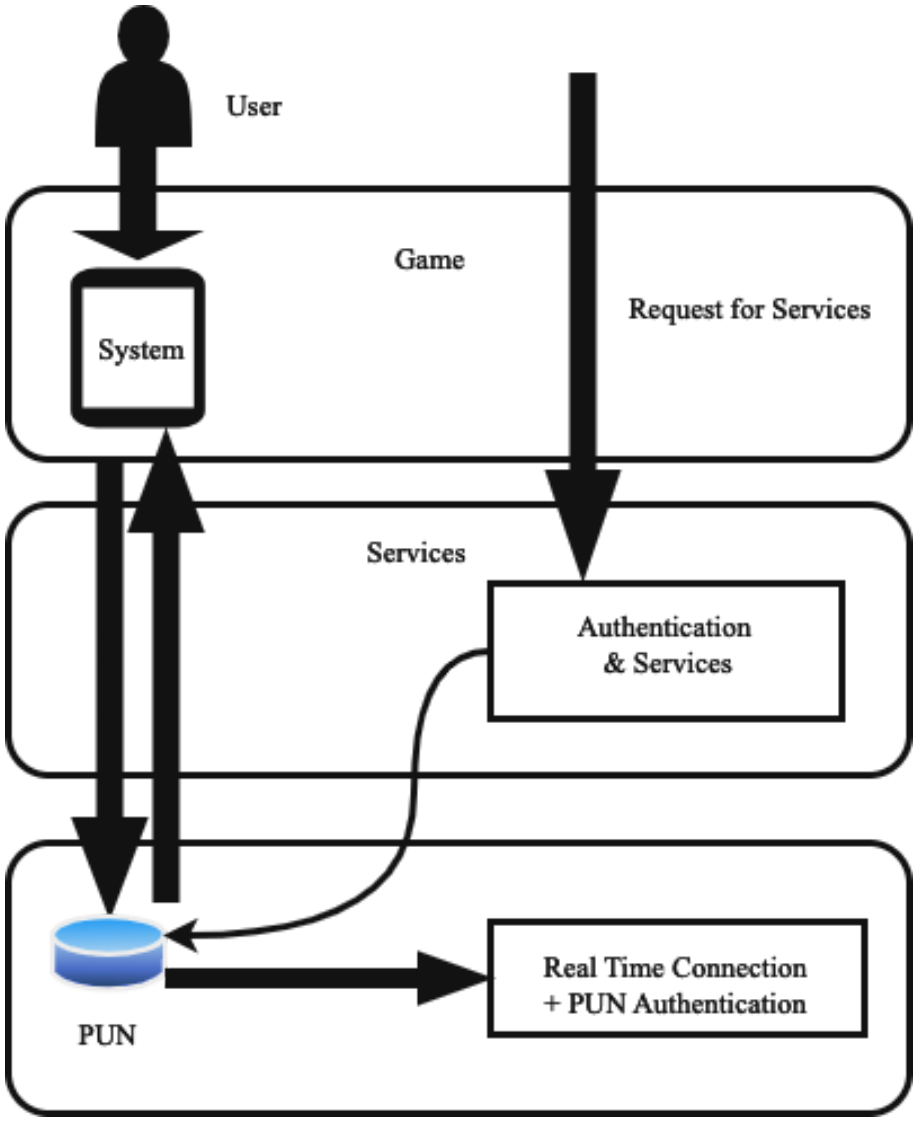


Figure 4.1 Architecture Diagram

4.2. Domain Model

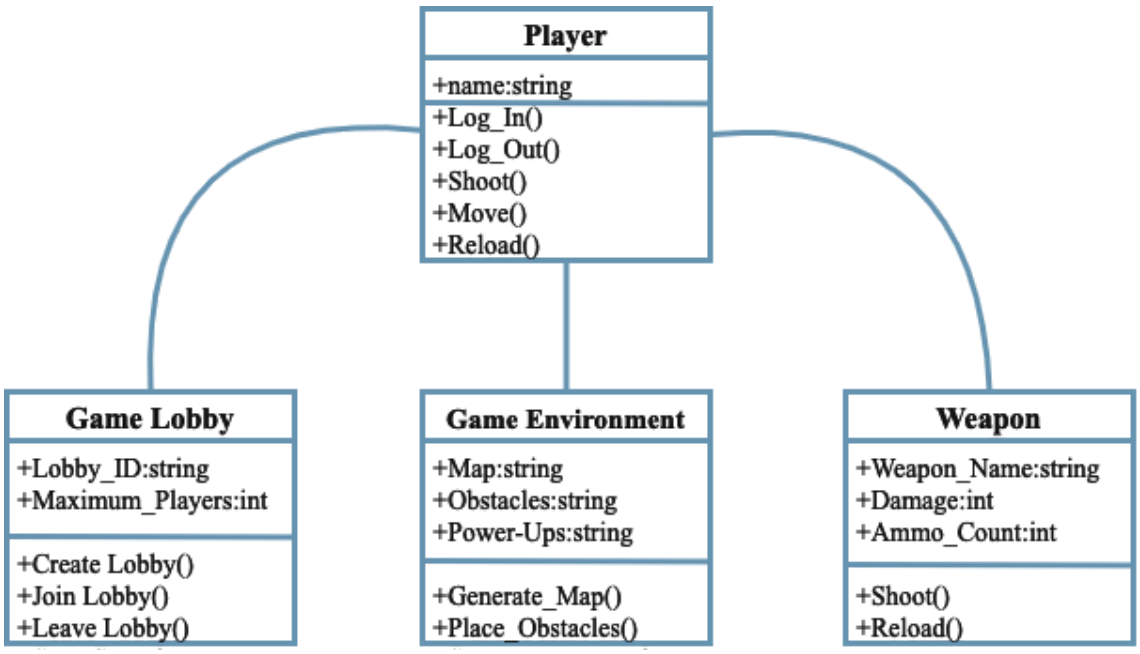


Figure 4.2 Domain Model

4.3. Entity Relationship Diagram

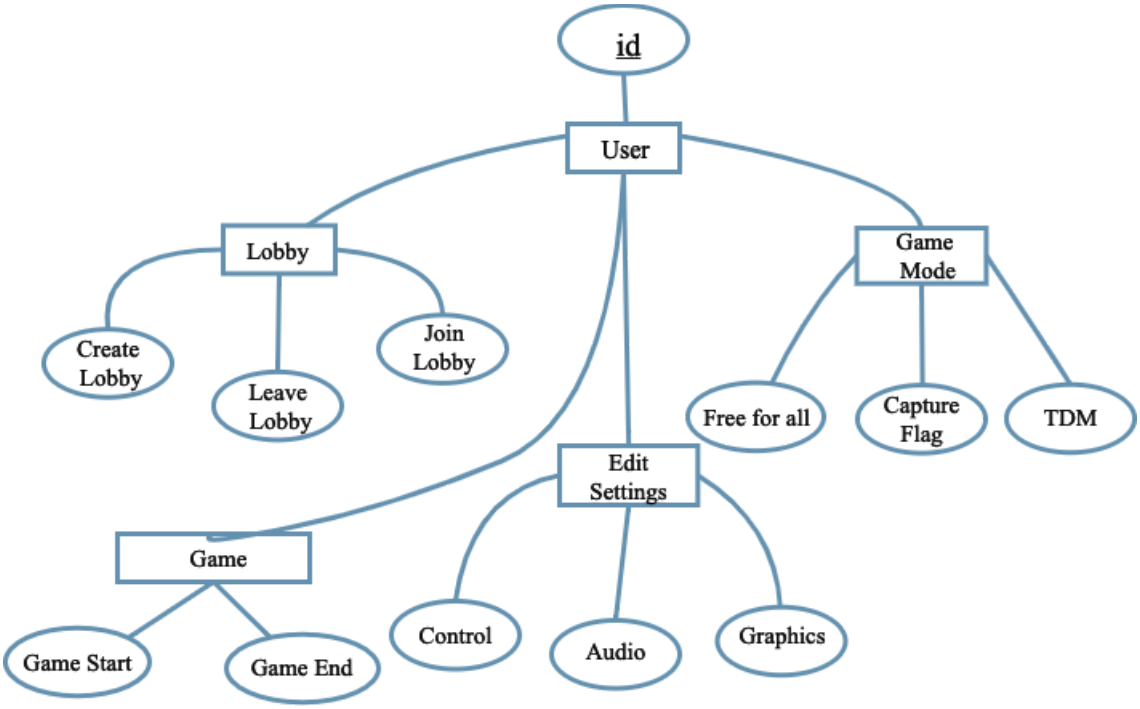


Figure 4.1 Entity Relationship Diagram

4.4. Class Diagram

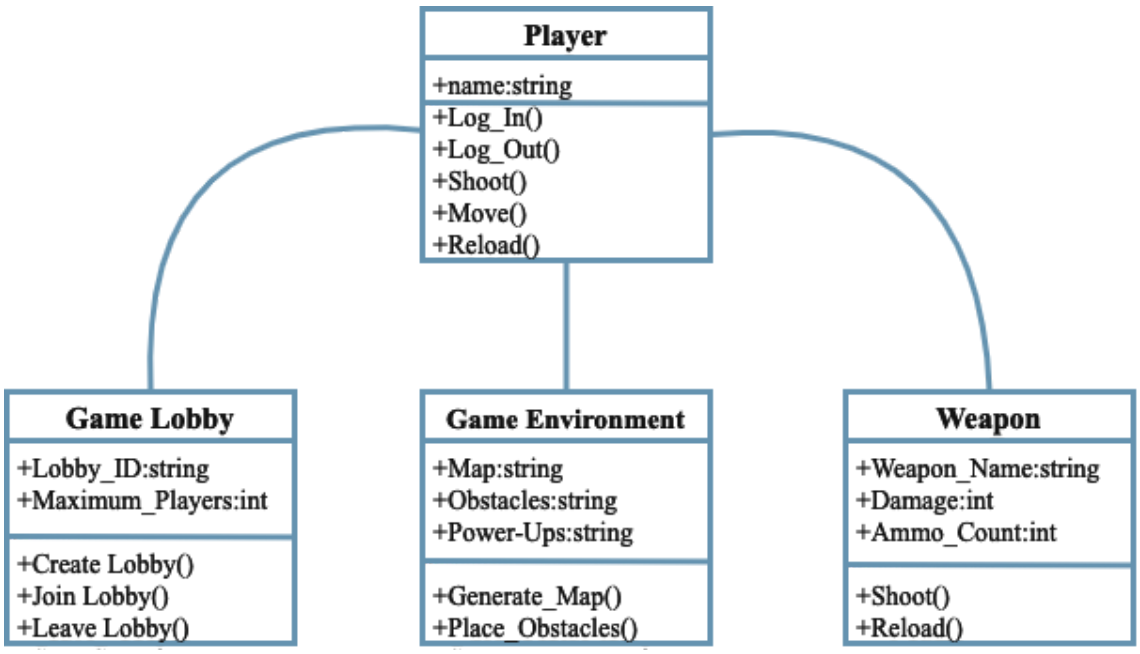


Figure 4.2 Class Diagram

4.5. Sequence / Collaboration Diagram

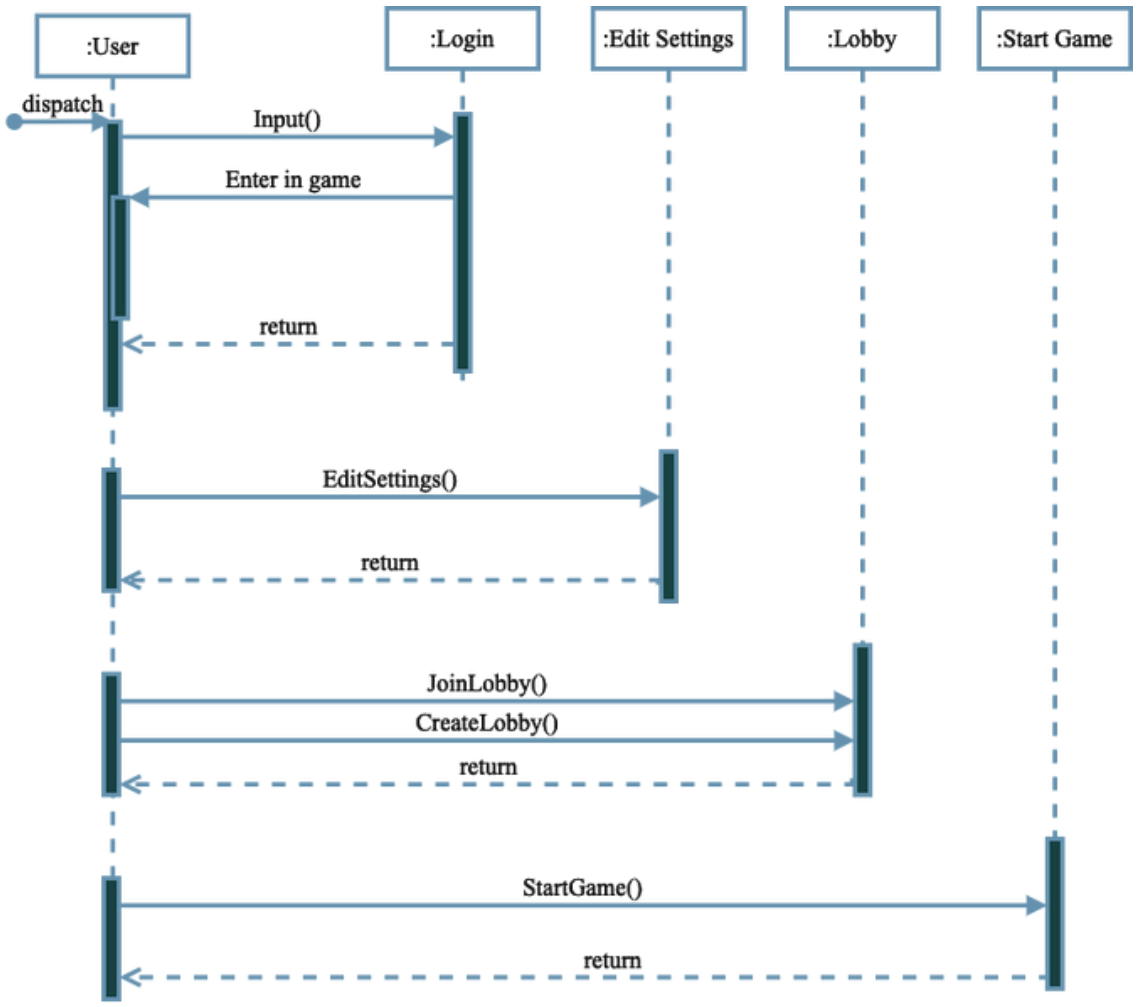


Figure 4.3 Sequence / Collaboration Diagram

4.6. Operation Contracts

- **Operation Contracts #1**

Table 4.1 Operation Contracts-01

Operation Name:	Login (username)
Cross Reference:	Use case Login
Pre-condition:	User should be online
Post-condition: -	User will be successfully log in and enter into the timeline **Objective:** User **Attribute:** (username) **Association:** the relationship between user and login

- **Operation Contracts #2**

Table 4.2 Operation Contracts-02

Operation Name:	Edit Settings (General, Graphics, Audio, Controls)
Cross Reference:	Use case Edit Settings
Pre-condition:	User should be online
Post-condition: -	User will be associated with profile (login) **Objective:** User **Attribute:** (General_Settings, Graphics_Settings, Audio_Settings, Controls_Settings) **Association:** relationship between user and Settings

- **Operation Contracts #3**

Table 4.3 Operation Contracts-03

Operation Name:	Join Lobby
Cross Reference:	Use case Join Lobby
Pre-condition:	User should be logged in and online
Post-condition: -	The Lobby joined randomly **Objective:** User **Attribute:** Free for all, Capture flags, Team death match **Association:** the relationship between user and lobby

- **Operation Contracts #4**

Table 4.4 Operation Contracts-04

Operation Name:	Create Lobby
Cross Reference:	Use case Create Game Lobby
Pre-condition:	User should be logged in and online
Post-condition: -	User created game lobby successfully **Objective:** User **Attribute:** (game mode, max players, room time limit, game goal, after finish, team selection, bots) **Association:** relationship between user and Lobby

- **Operation Contracts #5**

Table 4.5 Operation Contracts-05

Operation Name:	Start Game
Cross Reference:	Use case Start Game
Pre-condition:	User should be logged in and online
Post-condition: -	Game start successfully **Objective:** User **Attribute:** (move, shoot, reload, interact with object, check stats) **Association:** relationship between user and game

- **Operation Contracts#6**

Table 4.6 Operation Contracts-06

Operation Name:	Logout (user Id)
Cross Reference:	Use case Logout
Pre-condition:	User should be logged in and online
Post-condition: -	User will be logged out **Objective:** User **Attribute:** (user Id) **Association:** the relationship between user and Logout

4.7. Activity Diagram

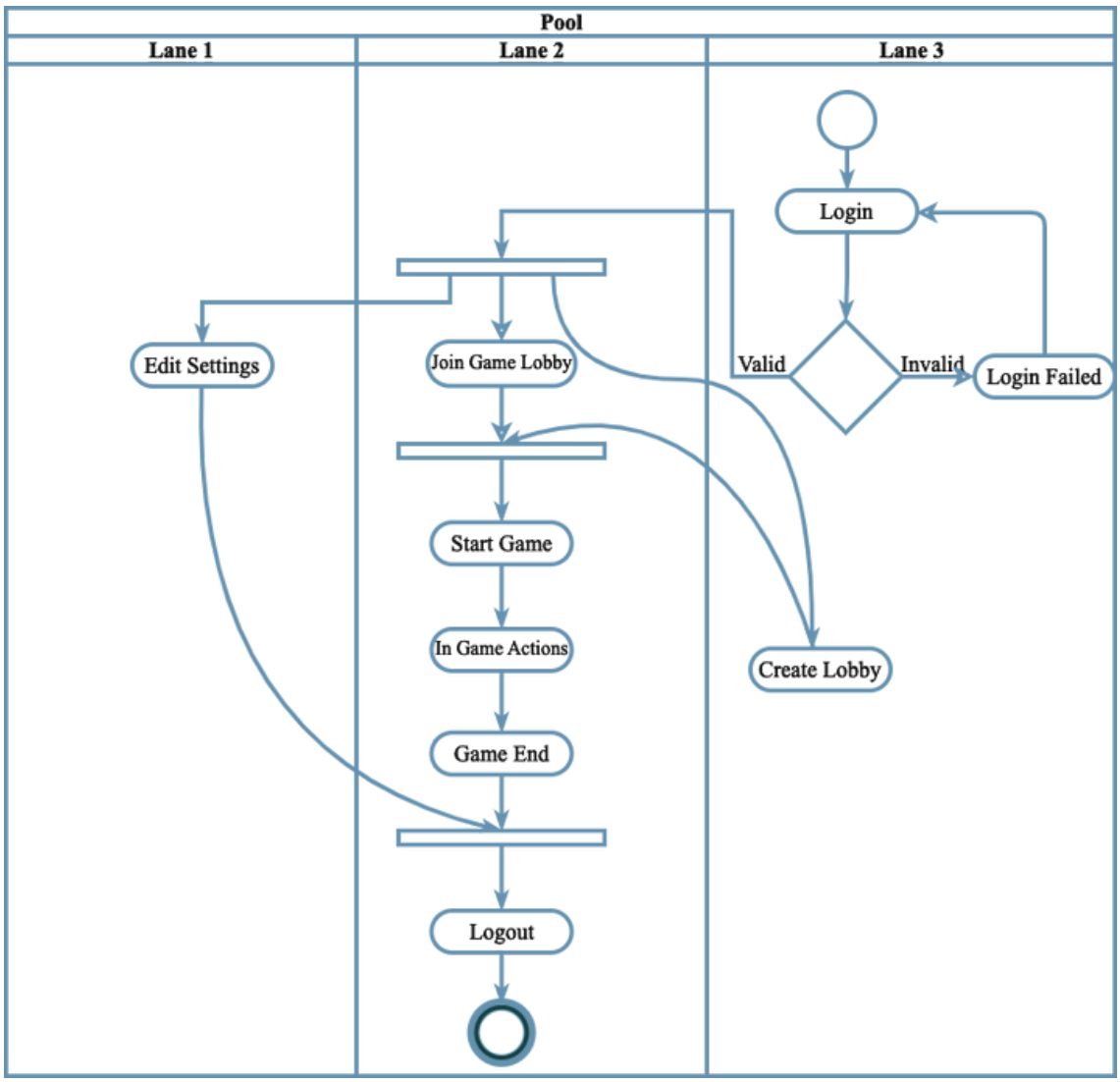


Figure 4.1 Activity Diagram

4.8. State Transition Diagram (Login)

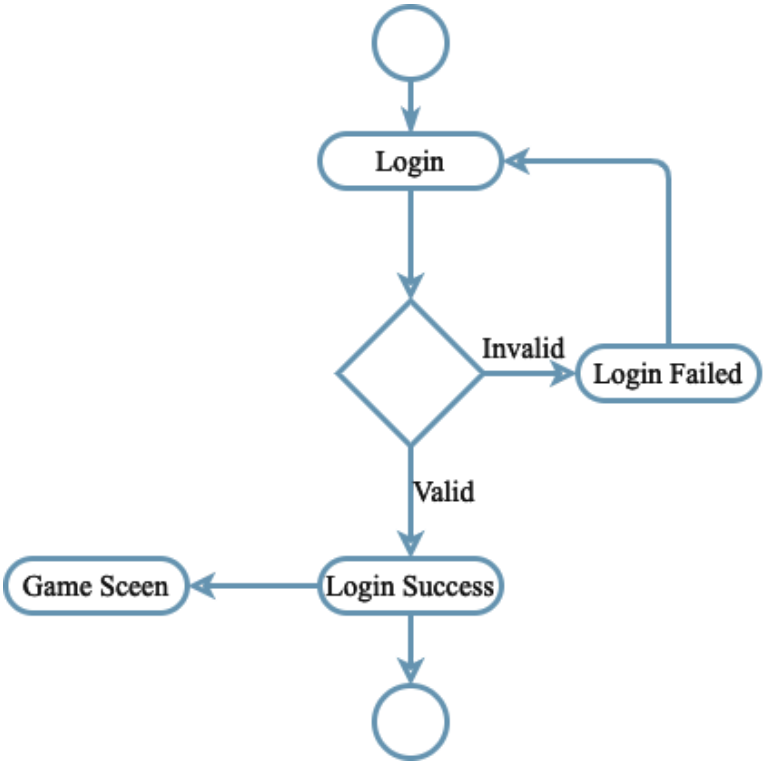


Figure 4.2 State Transition Diagram (Login)

4.9. State Transition Diagram (Edit Settings)

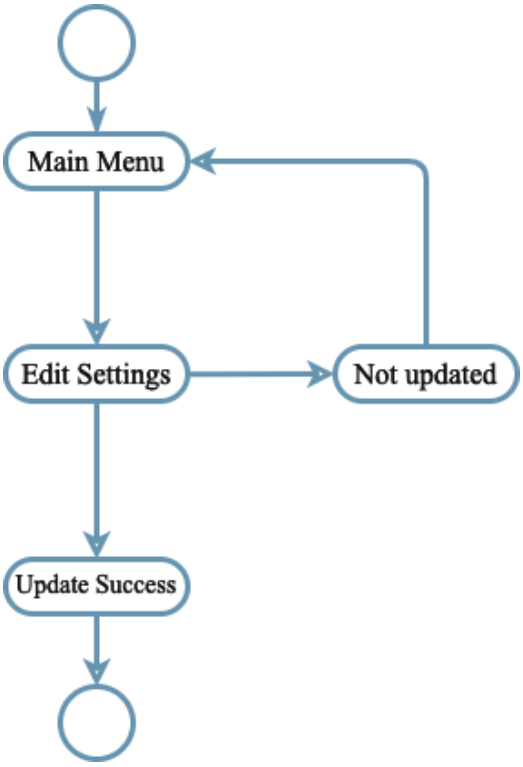


Figure 4.3 State Transition Diagram (Edit Settings)

4.10. State Transition Diagram (Join Game Lobby)

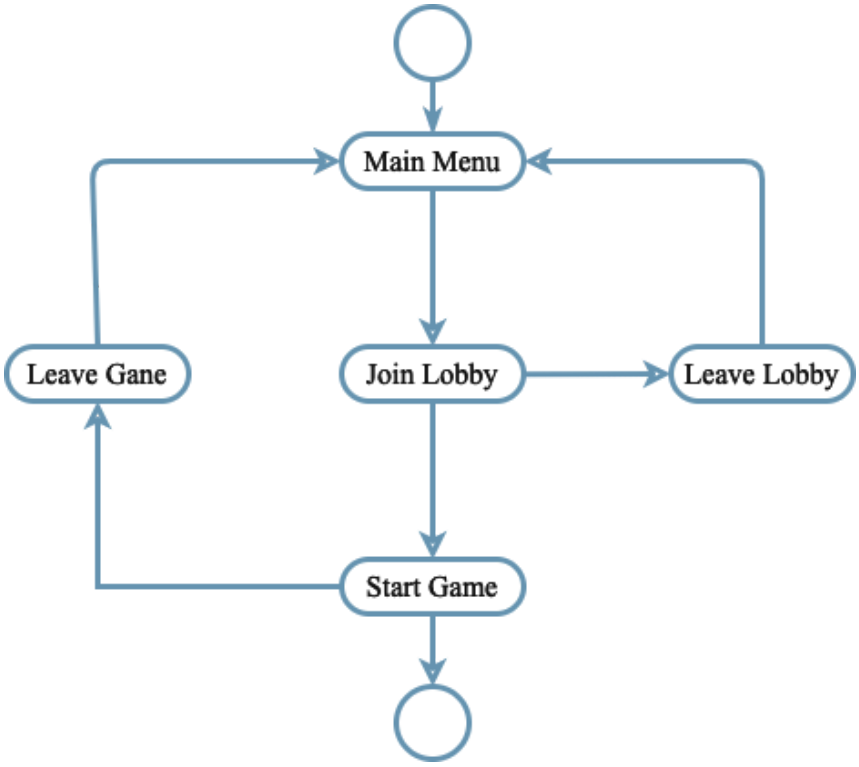


Figure 4.4 State Transition Diagram (Join Game Lobby)

4.11. State Transition Diagram (Create Lobby)

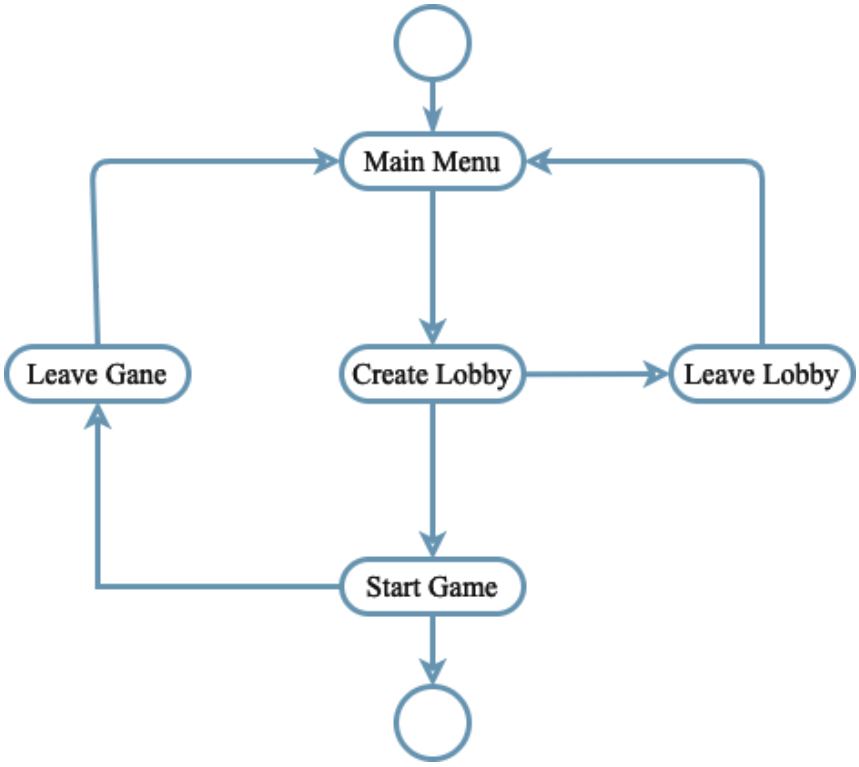


Figure 4.5 State Transition Diagram (Create Lobby)

4.12. State Transition Diagram (Start Game)



Figure 4.6 State Transition Diagram (Start Game)

4.13. Component Diagram

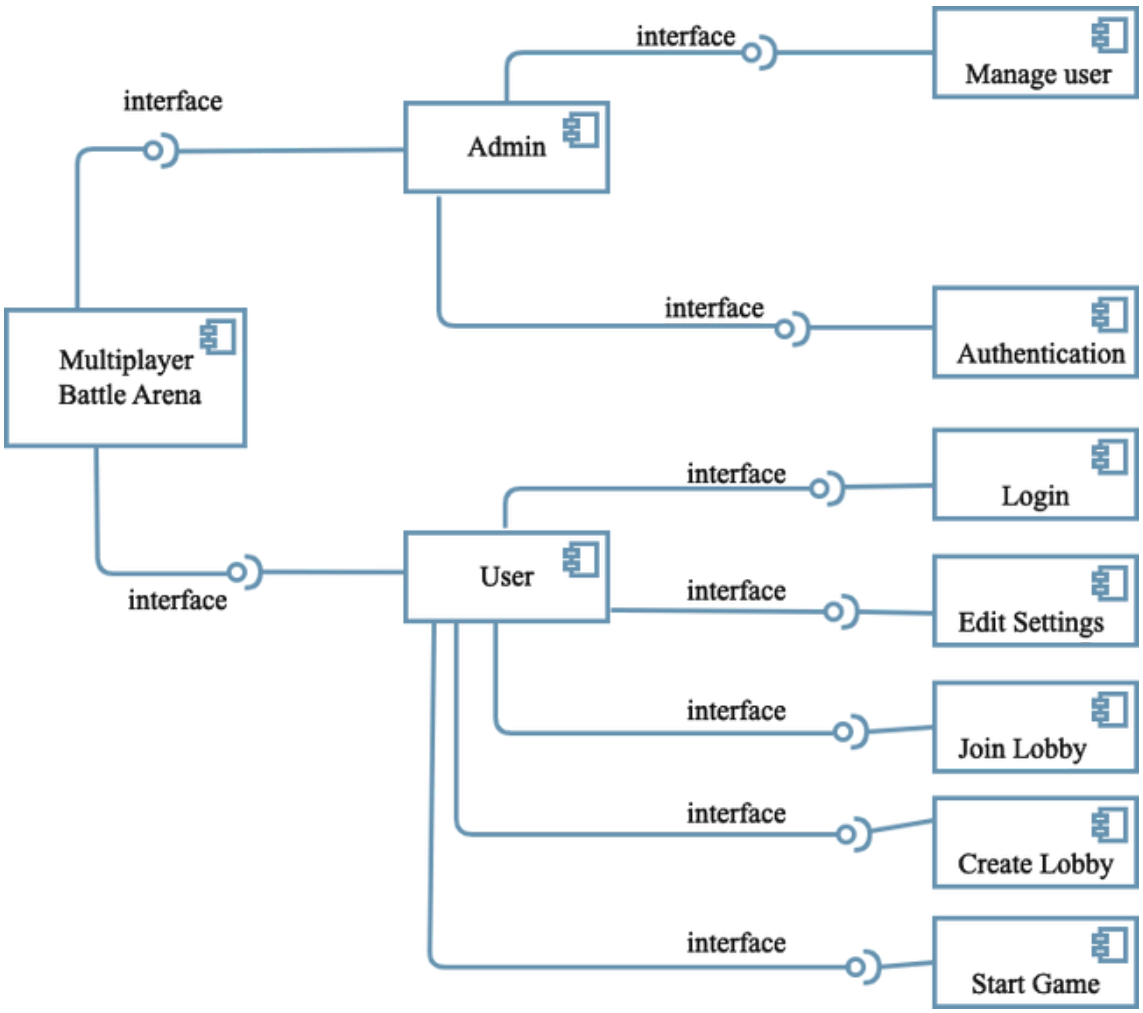


Figure 4.7 Component Diagram

4.14. Deployment Diagram

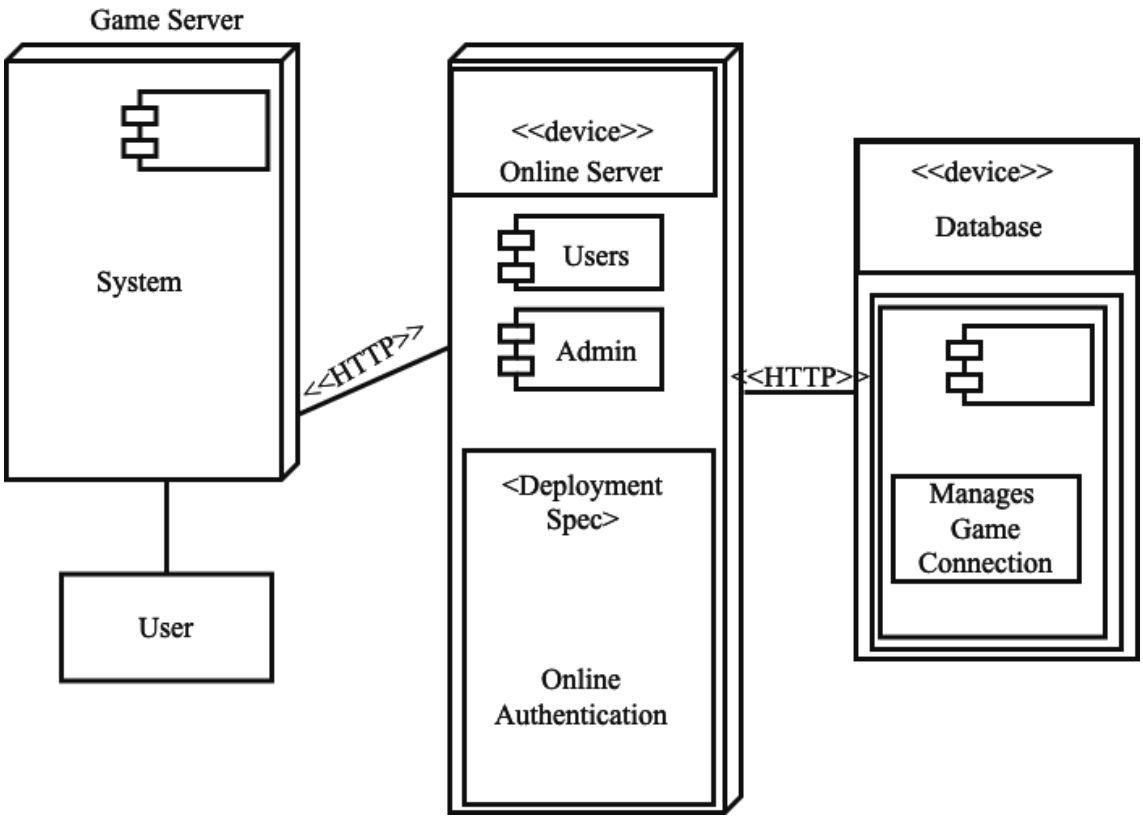


Figure 4.8 Deployment Diagram

4.15. Game Interference

4.15.1. Splash Screen

Splash Screen is the first loading screen which always show when the game opened, during this loading screen game connected to the server.



Figure 4.9 Splash Screen

4.15.2. Login Screen

After splash screen, login screen appear in which automatically guest name appears user can change the name or login with the given name and click on continue button.

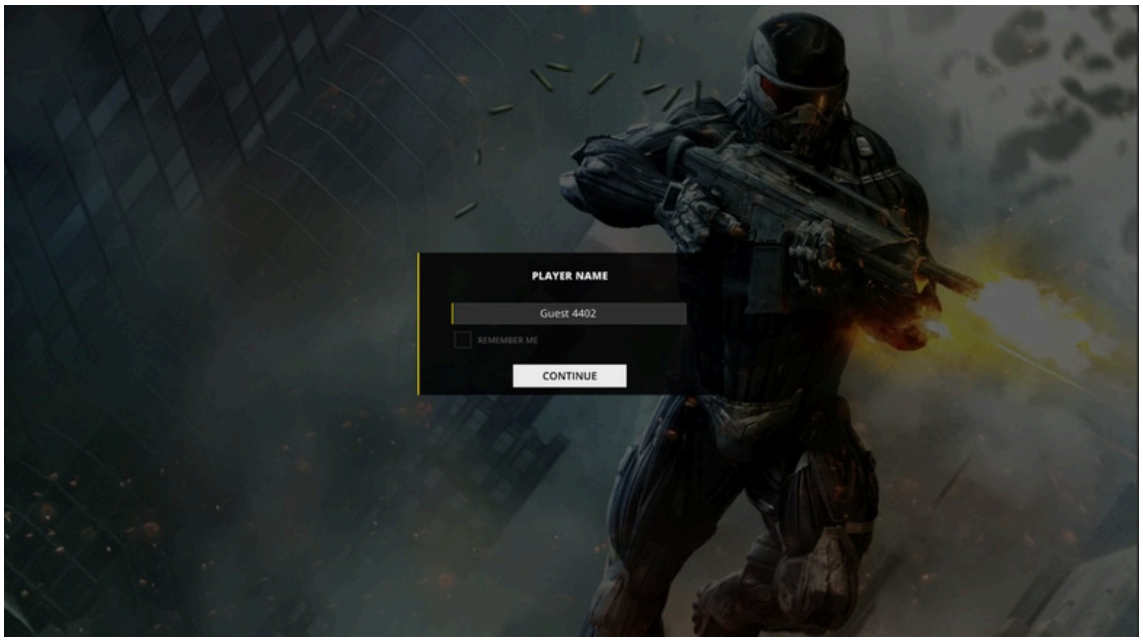


Figure 4.10 Login Screen

4.15.3. Main Menu Screen

After successfully login, main menu screen will be on screen. Through main menu screen user can choose multiple options like Play, options, logout or quit the game.

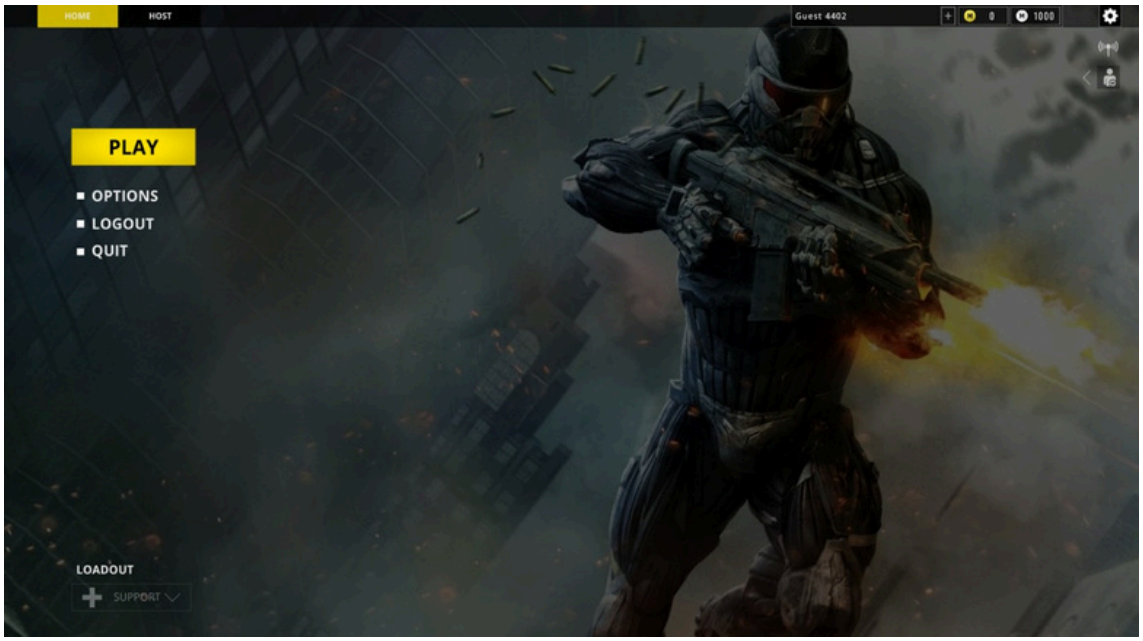


Figure 4.11 Main Menu Screen

4.15.4. General Settings

In general settings, user can change 3 things, show frame rate, refresh rate in seconds and weapon projection according to his needs.

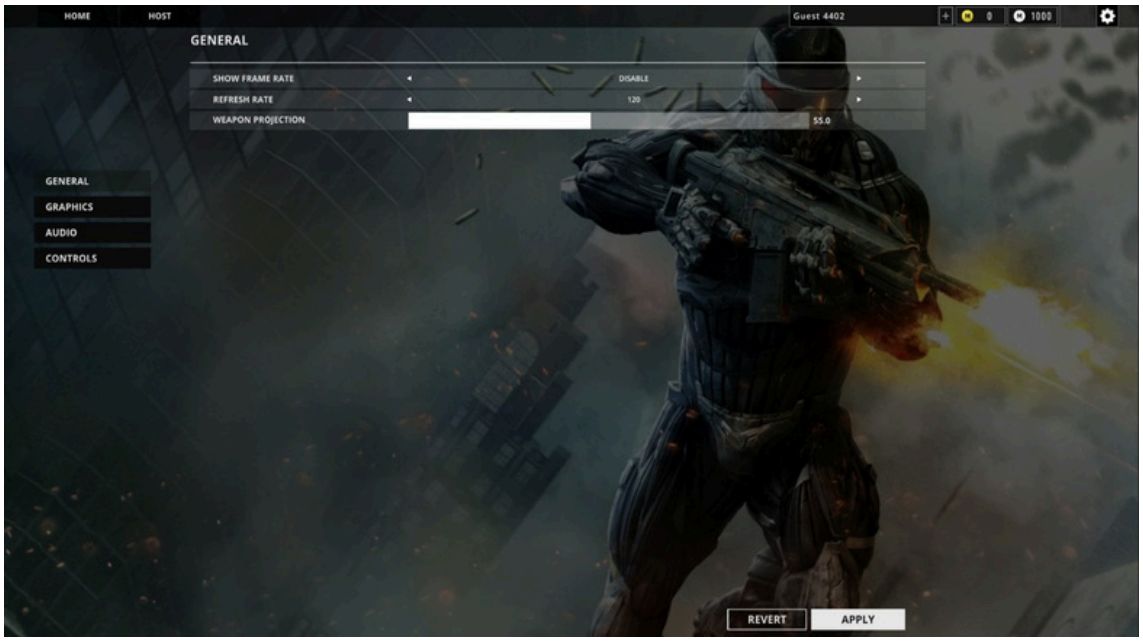


Figure 4.12 General Settings

4.15.5. Graphics Settings

In graphics settings, user can change the graphics settings according to his system requirements to run the game fast in his/her system.

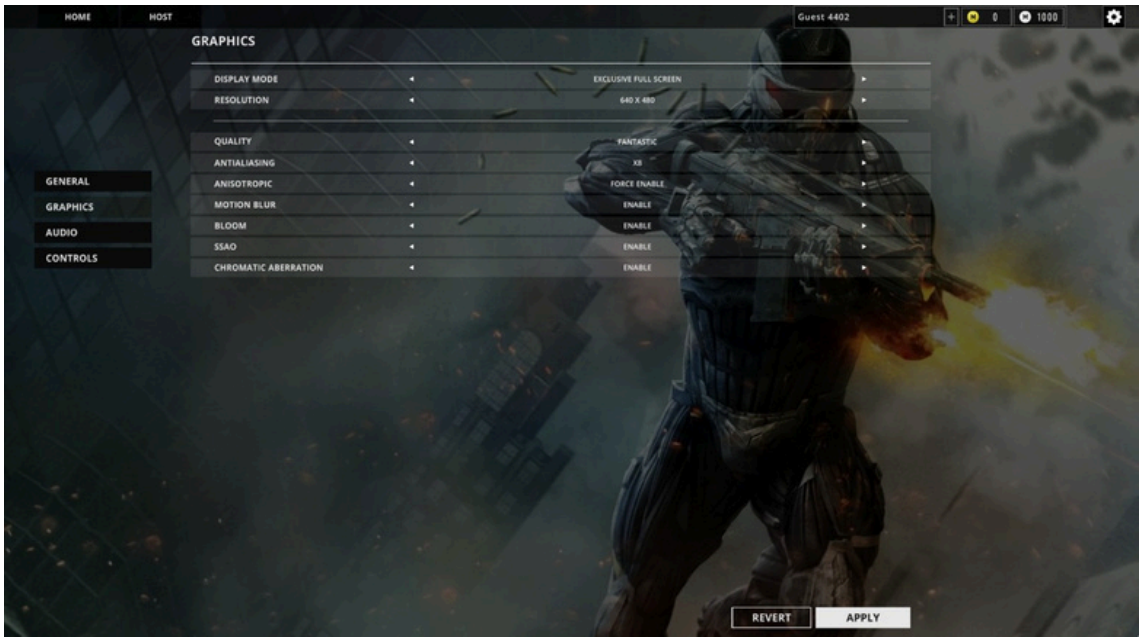


Figure 4.13 Graphics Settings

4.15.6. Audio Settings

In audio settings, user can increase and decrease the volume and music of the game according to his needs.

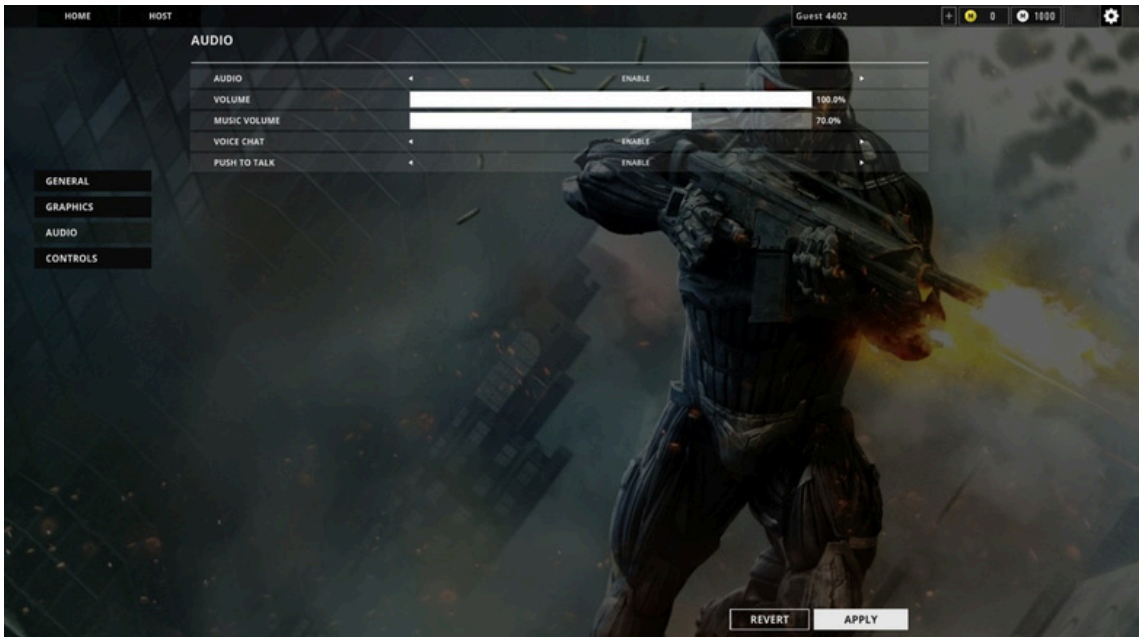


Figure 4.14 Audio Settings

4.15.7. Controls Settings

In controls settings, user can change the controls buttons that is suitable for him/user. After change must click on apply otherwise control settings should not be saved.

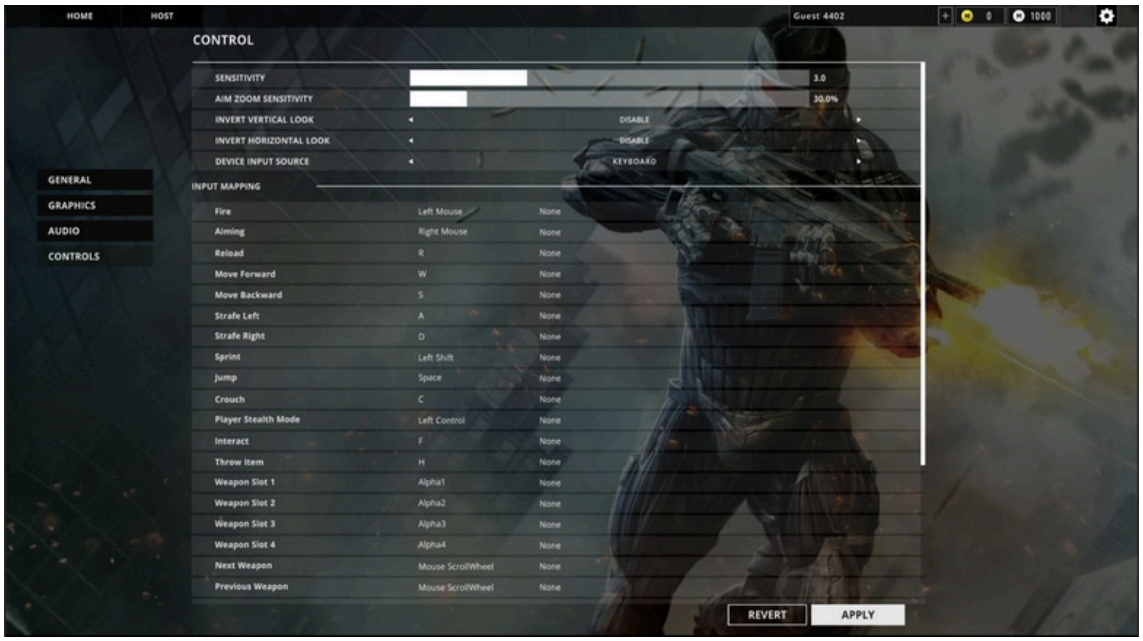


Figure 4.15 Controls Settings

4.15.8. Create Lobby Team Death Match Mode

Create lobby for the team death match must select the option of team death match while creating the lobby and click on create.

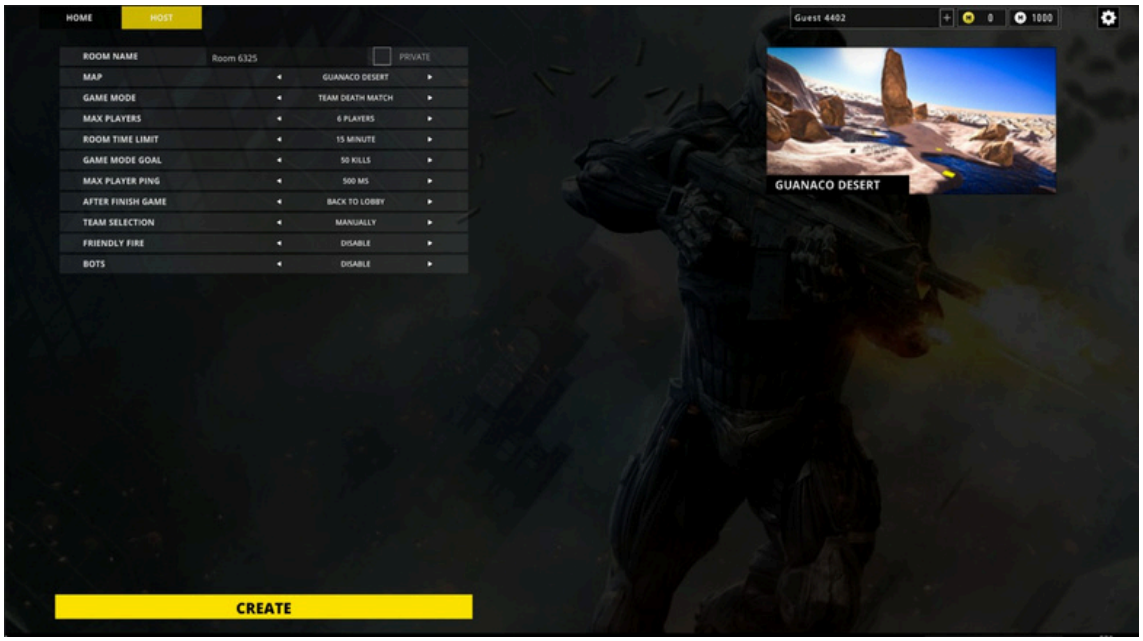


Figure 4.16 Create Lobby Team Death Match Mod

4.15.9. Create Lobby Capture of Flag Mode

Create lobby for the Capture of Flag must select the option of team death match while creating the lobby and click on create.

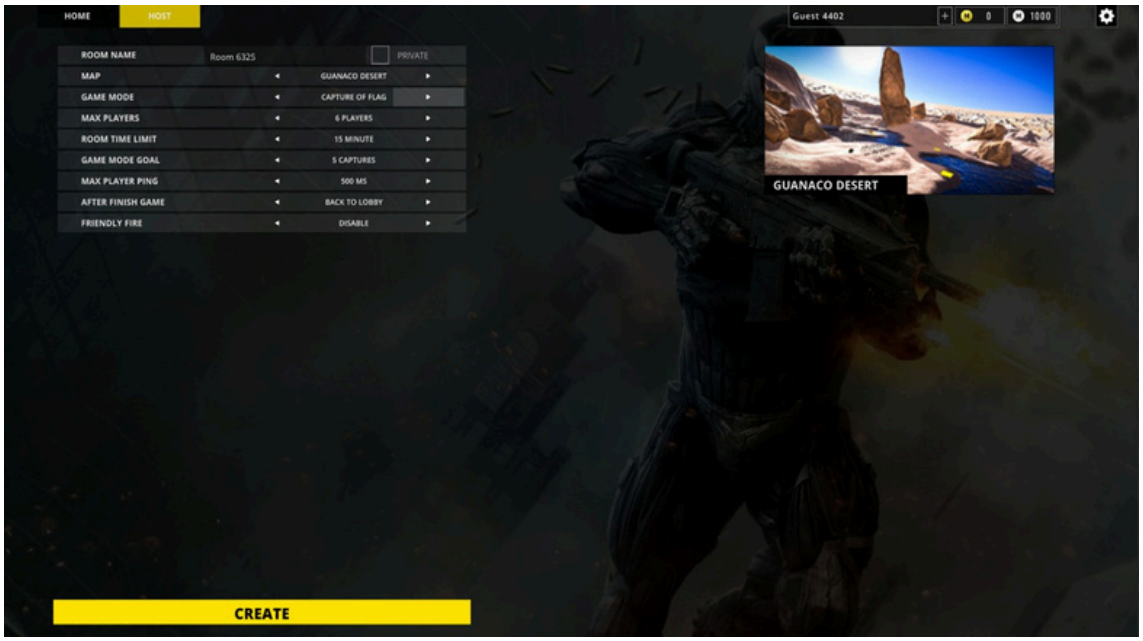


Figure 4.17 Create Lobby Capture Of Flags Mode

4.15.10. Create Lobby Free for All

Create lobby for the Free for all mode user must select the option of team death match while creating the lobby and click on create.

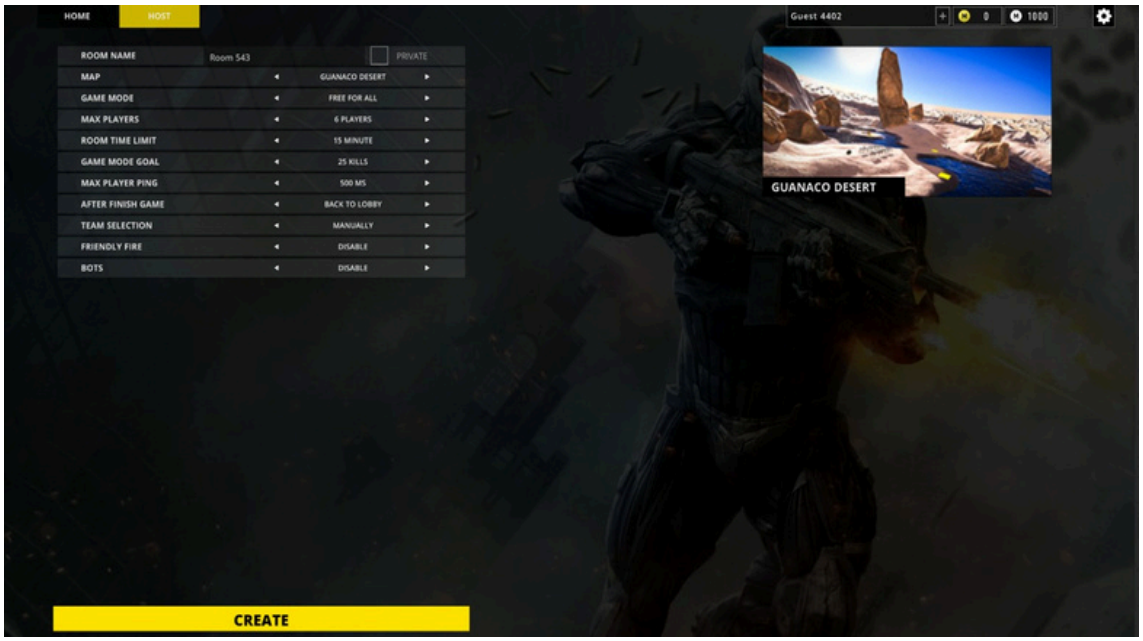


Figure 4.18 Create Lobby Free for All

4.15.11. Lobby

Start the game from the lobby, user have to click on start button to play the game.

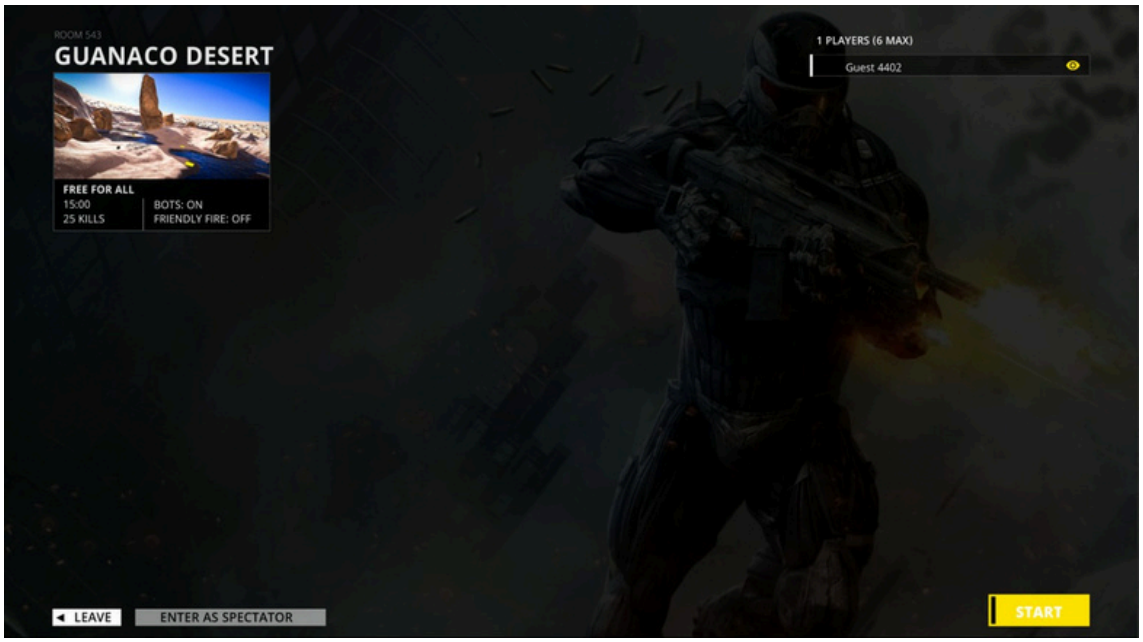


Figure 4.19 Lobby

4.15.12. Gameplay

Gameplay screen where user can play the game and kill other enemies to win the game. Here user can move, shoot, reload or exit the current game session.



Figure 4.20 Gameplay

4.15.13. Scoreboard

By click TAB button, user can see the scoreboard of the current gameplay session.



Figure 4.21 Scoreboard

CHAPTER 5

IMPLEMENTATION

5.1. Important Flow Control

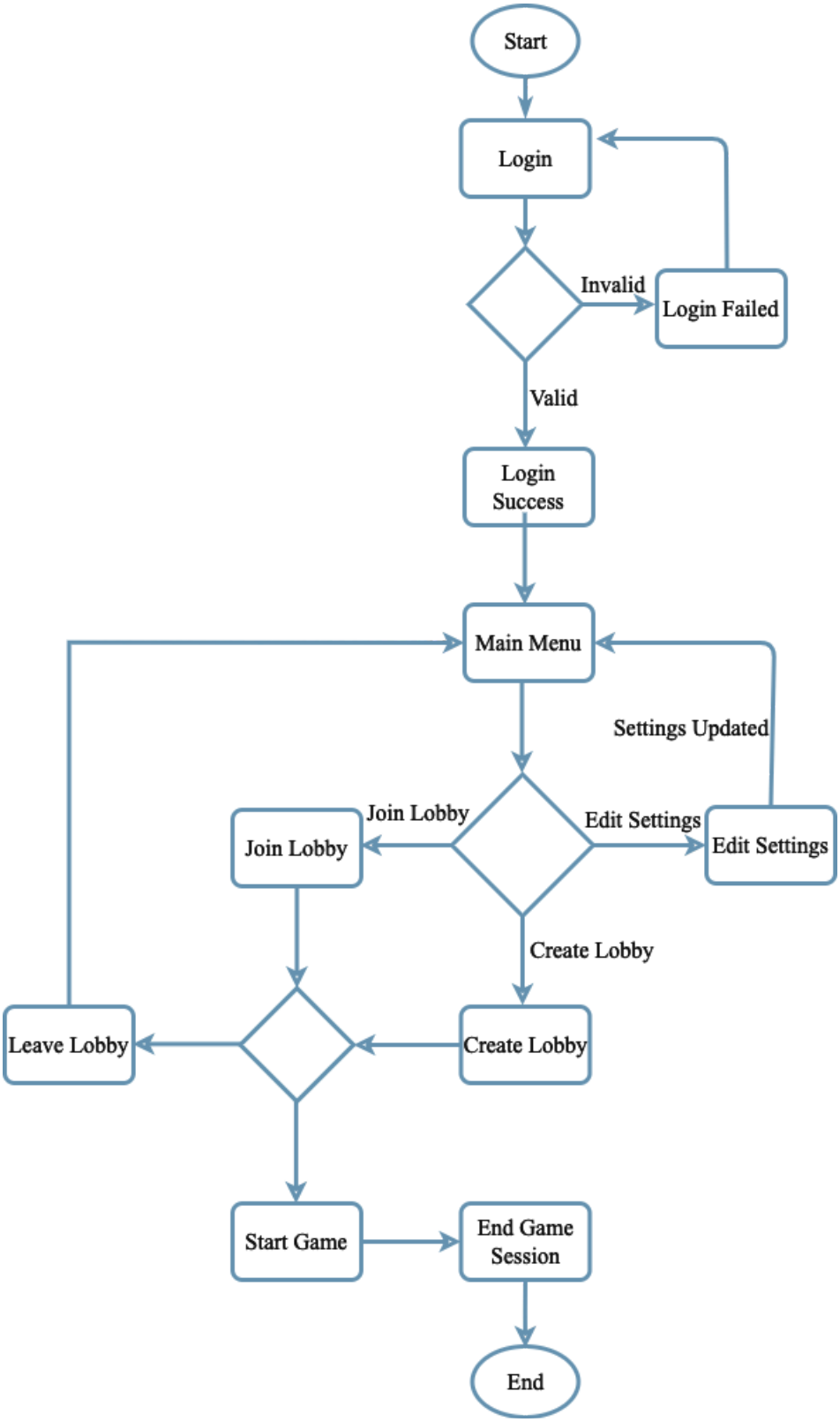


Figure 5.1 Important Flow Control

5.2. Components, Libraries and Stubs

Libraries

Libraries are a collection of diverse functions that are utilized in the creation of a project to make the various functions easier to perform. The functions are linked to the libraries, making it simple to conduct tasks across many activities. A list of the many libraries used in our project is provided below.

- Unity Libraries
- Photon Unity Networking

5.3. Deployment Environment

- Unity 3D
- C#
- Android Studio
- Visual Studio
- MacOS Montery

5.4. Tools and Techniques

- C#
- Android Studio
- Visual Studio
- Unity 3D
- Photon Unity Networking
- Blender
- Adobe Illustrator

5.5. Best Practices / Coding Standards

- Code should not have any trailing whitespace to avoid creating unnecessary diff issues.
- Function and classes are always commented on showing expected input and output
- GUI should be user-friendly
- Let photos say what you can’t say
- Document everything
- Demand for Feedback

CHAPTER 6

TEST AND EVALUATION

6.1. Testing and Evaluation

Testing is a procedure for detecting problems in a program. We execute a simulation test on the system to ensure that any errors or bugs are not there. Various scenarios and forms of testing will be used to determine whether or not the system is functioning properly. If any faults are discovered throughout the testing, provide a solution to the errors. In this chapter, we cover the entire system testing and assessment process, including how to use the system and what work each feature has completed.

6.2. Use Case Testing

Various scenarios, such as user and multiplayer functionality, are being tested throughout this session. Use Case Testing is a software testing technique that helps to identify test cases that cover entire system on a transaction-by-transaction basis from start to end. Test cases are the interactions between users and software application. Test **cases** based on use cases and are referred as scenarios. Capability to identify gaps in the system which would not be found by testing individual components in isolation. Very effective in defining the scope of acceptance tests.

- **Main Success Scenario**

Table 6.1 Use Case Testing-01

Actor: **User**		
No.	**Steps**	**Description**
1	Login	Login to the system with guest name System will redirect to next page
2	Edit Settings	Change the settings of the game according to your systems and your mindset
3	Create Game Lobby	Create game lobby according to your interest.
4	Join Lobby	Join any random lobby to play the game.
5	Start Game	Start the game-play and kill the enemies to win the game.
6	Logout	Logout success with press logout button System will redirect to login page

- **Extensions (Errors)**

Table 6.2 Use Case Testing Extension

No.	Step	Description
1-a	Login failed	Login failed if user did not have internet. System will display an error message
2-a	Lobby connection failed	Lobby connection failed if your internet connection is not stable
3-a	Lobby suddenly leave	Lobby suddenly leave if your internet connection lost during game
4-a	Lobby created failed	Lobby creation failed if your internet connection is not stable.
5-a	Game-play stucks	Game-play stucks if your internet connection lost.

6.3. Boundary Value Analysis

Boundary testing is a type of testing that takes place between a starting point and the extreme ends of a range of input values, or between partials of input values.

Login Username

- A person's username must be unique when entering it in the text field during login.

Text Fields

- The system must allow only correct text in all text fields.
- The user must add only the information that is required and brief.
- Name is examples of text fields.

6.4. Performance Testing

Test Case Title: System Response Time Testing

Table 6.3 Performance Testing 01

Test Type	Performance Testing	Test Result
Required performance	The required response time of the system is less than 10 seconds	
System performance	The actual performance response time is 5 seconds.	Test passed

Test Case Title: Activity Load Time Testing

Table 6.4 Performance Testing 02

Test Type	Performance Testing	Test Result
Required performance	The required page load time of the system is less than 5 seconds	
System performance	The actual page load time is 2 seconds.	Test passed

Test Case Title: Huge Number of Users

Table 6.5 Performance Testing 03

Test Type	Performance Testing	Test Result
Required performance	System should perform normally and efficiently with a large number of users	
System performance	System performed all activities normally with a large number of users.	Test passed

Test Case Title: Waiting Time Testing

Table 6.6 Performance Testing 04

Test Type	Performance Testing	Test Result
Required performance	The average waiting time of the system is less than 5 seconds	
System performance	The actual performance response time is 3 seconds in some circumstances.	Test passed

Test Case Title: Connection Establish Time Testing

Table 6.7 Performance Testing 05

Test Type	Performance Testing	Test Result
Required performance	The connection must be established in less than 3 seconds	
System performance	The actual connection establishes time is about 2 second.	Test passed

Test Case Title: Database Send/Get Information Time Testing

Table 6.8 Performance Testing 06

Test Type	Performance Testing	Test Result
Required performance	The system should save or retrieve data from database in less than 2 seconds.	
System performance	The actual time system sends or gets information to database in 1 second.	Test passed

6.5. Stress Testing

Stress testing is a method of determining a system's stability and dependability. Stress testing is a technique for determining a system's ability to handle flaws and faults. Stress testing is a technique for evaluating a system's ability to handle errors under harsh conditions. We overload our system and analyze its behavior in extreme settings to confirm that it behaves normally during stress. In bad conditions, the (Multiplayer Battle Arena) Game behaves regularly, and all of its features and functions work as expected. Through this testing, we monitor our system and make sure our system • System should save all data before crashing • Make sure the connection from server

- If server0 no response, wait on login current page

6.6. Improvement

As we all know, there is always room for improvement in every management system. The following are some basic changes that we should make in the future:

1. Add More Environments

To improve the game difficulty and enjoyment, need to add more environments that will be selectable by users. The more levels to get, the more user will play.

2. Game Quality

For use, quality is quite crucial. Always keep a close eye on quality control and never skimp on it. The higher the quality, the better.

3. System Level

Our system should not be clogged with too much data. We should only give the user with the most basic information. Otherwise, the user may feel perplexed and will be unable to use our game.

4. Add More Modes

Besides the 3 Modes, I am thinking to add more new mods in the multiplayer game to enjoy more in the game.

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